Homework 4

ORF 245 - Fundamentals of Statistics

Due: Feb. 28th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
- Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).

Exercise: 1. (Joint Discrete RV's) Let the pmf of (X, Y) be given by

	$p(x,y)$	y = 1	2	3	4
x	1	$\frac{1}{4}$	0	0	0
	2	$\frac{1}{8}$	$\frac{1}{8}$	0	0
	3	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	0
	4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

(a) Let A denote the event that $X = Y + 1$. Find $\mathbb{P}(A)$.

(b) Find the marginal pmf's of X, $p_X(x)$, and Y, $p_Y(y)$.

(c) Are the two random variables X and Y independent? Justify your answer.

(d) Find conditional pmf of X given $Y = 4$, $p_{X|Y}(x|4)$, and the conditional pmf of X given $Y = 1$, $p_{X|Y}(x|1)$.

Exercise: 2. (Joint Continuous RV's) Consider the random variables X, Y with the following joint probability density function:

$$f(x,y) = \begin{cases} 24xy, & 0 \le x \le 1,\ 0 \le y \le 1,\ x + y \le 1 \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

Note carefully the range of these random variables – both X and Y must be between $[0, 1]$, **and** their sum must be at most one.

(a) Show that the above satisfies the properties of a joint pdf.

(b) Find the marginal pdf of X, $f_X(x)$.

(c) Find the marginal pdf of Y, $f_Y(y)$.

(d) Are X and Y independent? Justify your answer.

(e) Find $\mathbb{E}[X]$ and $\mathbb{E}[Y]$.

Exercise: 3. (Properties of Jointly Distributed Random Variables)
Let X and Y be continuous random variables, with some joint density function $f(x, y)$.

(a) Let $f_Y(y)$ be the marginal pdf of Y. Show that for every value of y satisfying $f_Y(y) > 0$ we have:

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y)dx = 1.$$

Be sure to justify each step.

(b) Show that $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, *without using linearity of expectation.* That is, use only properties of integrals to show this result. Be sure to justify each step.

(Hint: Express the left-hand side as an integral, how can we simplify it or pull terms out?)

Exercise: 4. (Properties of Random Sample Mean)
Let$X_1, ..., X_n$ be independent and identically distributed (iid) continuous random variables. That is, their joint probability density function is equal to $f(x_1, ..., x_n) = f_X(x_1) \cdot \cdots \cdot f_X(x_n)$ for some common pdf $f_X(\cdot)$. Note that the marginal pdf of each $X_1, ..., X_n$ is given by the *same* function $f_X(\cdot)$.

Let the mean and variance of each X_i be given by $\mathbb{E}[X_1] = ... = \mathbb{E}[X_n] = \mu_X$, and $\text{Var}(X_1) = ... = \text{Var}(X_n) = \sigma_X^2$, for some values μ_X and $\sigma_X^2 > 0$.

(a) Show that $\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \mu_X$. Be sure to justify each step.

(b) Show that $\text{Var}\left(\frac{1}{n}\sum_{i=1}^n X_i\right) = \frac{\sigma_X^2}{n}$. Be sure to justify each step.

(c) Provide a one-sentence interpretation of the result in part (b) – what happens to the distribution of $\frac{1}{n}\sum_{i=1}^n X_i$ as n increases, i.e. as we obtain more and more independent random variables coming from the same distribution?

Exercise: 5. [OPTIONAL] (Joint Contnuous RV's II)
Note: This is an ***optional*** exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Suppose random variables X and Y have the following joint PDF:

$$f(x, y) = \begin{cases} 1, & 0 \leq x \leq 1 \quad \text{and} \quad x \leq y \leq x + 1, \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the marginal distribution of X, $f_X(x)$. What is the name of this family of distributions? What are the associated parameters that characterize the marginal pdf of X?

(b) Compute $\mu_X = \mathbb{E}[X]$ and $\sigma_X^2 = \text{Var}(X)$.

(c) Find the marginal distribution of Y, $f_Y(y)$.

(Hint: Break up the marginal distribution into two parts. First find the marginal distribution of Y for $0 \leq y \leq 1$, and then for $1 < y \leq 2$. Think carefully about the limits of integration – a 2-D diagram of the range of X and Y could be useful.)

(d) What is $\mu_Y = \mathbb{E}[Y]$?

(e) Find $\text{Cov}(X, Y)$, the covariance between X and Y.

(f) Are X and Y independent? Justify your answer.